INNOVATION PROJECT ABSTRACTS

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1. English Abstract: Integrated Waste Management & Route Optimization System

Rapid urbanization across Kenyan municipalities has created critical inefficiencies in solid waste collection, request tracking, and logistical routing. This technical innovation project presents a full-stack Waste Management System engineered using HTML5, CSS3, JavaScript, and React, paired with a MySQL relational database and a Travelling Salesman Problem (TSP) algorithmic solver. Developed collaboratively within a five-member engineering team, the platform features secure user authentication, automated password recovery, and Role-Based Access Control (RBAC). Citizens can log real-time waste collection requests and view municipal services, while administrators monitor system-wide operations, manage user accounts, and utilize TSP route optimization to minimize collection vehicle fuel consumption and turnaround times. This innovation demonstrates how modern web architectures and classical algorithms can be synthesized to resolve pressing environmental and infrastructural challenges.

2. Kiswahili Abstract: Mfumo Jumuishi wa Usimamizi wa Taka na Uboreshaji wa Njia

Ukuaji wa haraka wa miji nchini Kenya umesababisha changamoto kubwa katika ukusanyaji wa taka ngumu, ufuatiliaji wa maombi ya huduma, na upangaji wa njia za usafirishaji. Mradi huu wa ubunifu wa kiufundi unawasilisha Mfumo wa Usimamizi wa Taka ulioundwa kwa kutumia HTML5, CSS3, JavaScript, na React, pamoja na hifadhidata ya MySQL na algoriti ya Travelling Salesman Problem (TSP). Mfumo huu ulitengenezwa kwa ushirikiano wa timu ya watu watano na unajumuisha uthibitishaji salama wa watumiaji, urejeshaji wa nywila, na udhibiti wa ufikiaji kulingana na majukumu (RBAC). Wakazi wanaweza kuwasilisha maombi ya ukusanyaji wa taka moja kwa moja mtandaoni, huku wasimamizi wakifuatilia shughuli zote na kutumia suluhisho la TSP kupunguza gharama za mafuta na muda wa safari. Ubunifu huu unaonyesha jinsi teknolojia ya wavuti inavyoweza kutatua changamoto za mazingira mijini.

3. Indigenous (Kikamba) Abstract: Mũvango wa Kũsũvĩa Taka na Kũseũvya Nzĩa sya Mawĩra

Kũkũa kwa mĩsyĩ na ndũa nthĩnĩ wa Kenya nĩkũetete mathĩna manene ma kũkusany'a taka, kũatĩĩana na maombi ma andũ, na kũvanga nzĩa sya ngalĩ sya kũua taka. Wĩra ũũ wa ũũĩ wa tekinolojia wĩonany'a Mũvango wa Kũsũvĩa Taka ũla ũseũvĩtw'e kwa kũtũmĩa HTML5, CSS3, JavaScript, na React, vamwe na MySQL na programu ya Travelling Salesman Problem (TSP). Mũvango ũũ waseũviwe nĩ kĩkundi kya amanyĩw'a atano na wĩna nzĩa sya usalama sya kwĩlikya mtandaonĩ na kũtũngĩĩa nywila (password recovery). Atũmĩi nĩmatonya kũtũma maombi ma kũvewa ũtethyo wa kũvetangwa taka, nao atongoi (administrators) makasũvĩa maombi asu na kũtũmĩa nzĩa ya TSP kũolanga ivinda na mavuta ma ngalĩ. Wĩra ũũ wĩonany'a ũndũ tekinolojia ya mĩthenya ĩĩ ĩtonya kũtũmĩka kũseũvya mawĩthyũlũlũko maitũ.